

概率论第三章作业

1. 一袋内装有 5 个白球, 3 个红球。第一次从袋中任意取一个球, 不放回; 第二次又从袋中任取两个球, X_i 表示第 i 次取到的白球数, $i = 1, 2$ 。求:

(a) (X_1, X_2) 的分布及边缘分布;

(b) $P\{X_1 = 0, X_2 \neq 0\}$, $P\{X_1 = X_2\}$, $P\{X_1 X_2 = 0\}$

解:

(a)

$$P\{X_1 = 0, X_2 = 0\} = \frac{3}{8} \times \frac{1}{C_7^2} = \frac{1}{56}$$

$$P\{X_1 = 0, X_2 = 1\} = \frac{3}{8} \times \frac{C_5^1 C_2^1}{C_7^2} = \frac{5}{28}$$

$$P\{X_1 = 0, X_2 = 2\} = \frac{3}{8} \times \frac{C_5^2}{C_7^2} = \frac{5}{28}$$

$$P\{X_1 = 1, X_2 = 0\} = \frac{5}{8} \times \frac{C_3^2}{C_7^2} = \frac{5}{56}$$

$$P\{X_1 = 1, X_2 = 1\} = \frac{5}{8} \times \frac{C_4^1 C_3^1}{C_7^2} = \frac{5}{14}$$

$$P\{X_1 = 1, X_2 = 2\} = \frac{5}{8} \times \frac{C_4^2}{C_7^2} = \frac{5}{28}$$

联合概率分布

x	$f(x_2)$	0	1
x_1	$\frac{1}{56}$	$\frac{5}{28}$	$\frac{5}{28}$
1	$\frac{5}{56}$	$\frac{5}{14}$	$\frac{5}{28}$

边缘概率分布

$$P\{X_1 = 0\} = \frac{1}{56} + \frac{5}{28} + \frac{5}{28} = \frac{21}{56} = \frac{3}{8}$$

$$P\{X_1 = 1\} = \frac{5}{56} + \frac{5}{14} + \frac{5}{28} = \frac{35}{56} = \frac{5}{8}$$

$$P\{X_2 = 0\} = \frac{1}{56} + \frac{5}{56} = \frac{3}{28}$$

$$P\{X_2 = 1\} = \frac{5}{28} + \frac{5}{14} = \frac{15}{28}$$

$$P\{X_2 = 2\} = \frac{5}{28} + \frac{5}{28} = \frac{5}{14}$$

即 X_1 的边缘概率分布为

x_1	0	1
p	$\frac{3}{8}$	$\frac{5}{8}$

X_2 的边缘概率分布为

x_i	0	1	2
p	$\frac{3}{28}$	$\frac{15}{28}$	$\frac{5}{14}$

(b)

$$\begin{aligned}
 P\{X_1 = 0, X_2 \neq 0\} &= P\{X_1 = 0, X_2 = 1\} + P\{X_1 = 0, X_2 = 2\} \\
 &= \frac{5}{28} + \frac{5}{28} = \frac{5}{14} \\
 P\{X_1 = X_2\} &= P\{X_1 = 0, X_2 = 0\} + P\{X_1 = 1, X_2 = 1\} \\
 &= \frac{1}{56} + \frac{5}{14} = \frac{3}{8} \\
 P\{X_1 X_2 = 0\} &= P\{X_1 = 0, X_2 = 0\} + P\{X_1 = 0, X_2 = 1\} \\
 &\quad + P\{X_1 = 0, X_2 = 2\} + P\{X_1 = 1, X_2 = 0\} \\
 &= P\{X_1 = 0\} + P\{X_1 = 1, X_2 = 0\} \\
 &= \frac{3}{8} + \frac{5}{56} = \frac{13}{28}
 \end{aligned}$$

2. 已知二维随机变量 (X, Y) 的联合分布律为:

$\begin{matrix} & Y \\ X \end{matrix}$	2	4	6
1	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{12}$
3	$\frac{2}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

求:

(a) $Z = X + Y$ 的分布律;

(b) $M = \max(X, Y)$ 的分布律;

(c) $N = \min(X, Y)$ 的分布律;

解:

(a)

$$\begin{aligned}
 P(Z = 3) &= p_{12} = \frac{1}{6}, \quad P\{Z = 5\} = p_{11} + p_{33} = \frac{5}{12} \\
 P\{Z = 7\} &= p_{16} + p_{34} = \frac{1}{4}, \quad P\{Z = 9\} = p_{35} = \frac{1}{6}
 \end{aligned}$$

因此 $Z = X + Y$

Z	3	5	7	9
p_k	$\frac{1}{6}$	$\frac{5}{12}$	$\frac{1}{4}$	$\frac{1}{6}$

(b)

$$\begin{aligned}
 P\{M = 2\} &= p_{12} = \frac{1}{6}, \quad P\{M = 3\} = p_{32} = \frac{1}{3} \\
 P\{M = 4\} &= p_{14} + p_{34} = \frac{1}{4}, \quad P\{M = 6\} = p_{16} + p_{36} = \frac{1}{4}
 \end{aligned}$$

所以 $M = \max(X, Y)$ 的分布律为下表:

M	2	3	4	6
p_k	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{4}$

(c)

$$P\{N=1\}=p_{12}+p_{14}+p_{16}=\frac{1}{3}, \quad P\{N=2\}=p_{32}=\frac{1}{3}$$
$$P\{N=3\}=p_{34}+p_{36}=\frac{1}{3}$$

所以 $N=\min(X,Y)$ 的分布律为下表:

N	1	2	3
p_k	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

3. 设二维随机变量 (X,Y) 的联合概率密度函数为:

$$f(x,y)=\begin{cases} \frac{1+xy}{4}, & |x|<1, |y|<1 \\ 0, & \text{其他} \end{cases}$$

证明 X 与 Y 不独立, 但 X^2 与 Y^2 独立。

(a) 证对 X,Y 而言:

$$f_X(x)=\begin{cases} \frac{1}{2}, & |x|<1 \\ 0, & \text{其他} \end{cases} \quad f_Y(y)=\begin{cases} \frac{1}{2}, & |y|<1 \\ 0, & \text{其他} \end{cases}$$

因为 $f(x,y) \neq f_X(x)f_Y(y)$, 所以 X,Y 不独立

$$F_U(u)=P\{X^2 \leq u\}=\begin{cases} 0, & u<0 \\ \sqrt{u}, & 0 \leq u<1 \\ 1, & u \geq 1 \end{cases}$$
$$F_V(v)=P\{Y^2 \leq v\}=\begin{cases} 0, & v<0 \\ v, & 0 \leq v<1 \\ 1, & v \geq 1 \end{cases}$$

$U=X^2, V=Y^2$ 的联合分布函数为

$$F(u,v)=P\{X^2 \leq u, Y^2 \leq v\}=\begin{cases} 0, & u<0 \text{ 或 } v<0 \\ \sqrt{uv}, & 0 \leq u<1, 0 \leq v<1 \\ \sqrt{u}, & 0 \leq u<1, 1 \leq v \\ \sqrt{v}, & 1 \leq u, 0 \leq v<1 \\ 1, & 1 \leq u, 1 \leq v \end{cases}$$

可见对 $U=X^2, V=Y^2$ 而言, 有 $F(u,v)=F_U(u)F_V(v)$ 即 Y^2 和 X^2 相互独立

4. 设随机变量 (X,Y) 的联合密度函数为:

$$f(x,y)=\begin{cases} 3x, & 0<x<1, 0<y<x \\ 0, & \text{其他} \end{cases}$$

求 $Z=X+Y$ 的密度函数 $f_Z(z)$

解:

(a)

$$f_z(z) = P(Z \leq z) = P(X + Y \leq z) = \iint_{x+y \leq z} f(x, y) dx dy$$

当 $z < 0$ 时

$$F_z(z) = 0$$

当 $0 \leq z < 1$ 时

$$\begin{aligned} F_2(z) &= \iint_{x+y \leq z} f(x, y) dx dy \\ &= \int_0^{\frac{z}{2}} dy \int_y^{z-y} 3x dx \\ &= \int_0^{\frac{z}{2}} \left(\frac{3}{2} z^2 - 3zy \right) dy \\ &= \frac{3}{4} z^3 - \frac{3}{8} z^3 = \frac{3}{8} z^3 \end{aligned}$$

当 $1 \leq z < 2$ 时

$$\begin{aligned} F_z(z) &= \int_0^{\frac{z}{2}} dx \int_0^x 3x dy + \int_{\frac{z}{2}}^1 dx \int_0^{z-x} 3x dy \\ &= \int_0^{\frac{z}{2}} 3x^2 dx + \int_{\frac{z}{2}}^1 (3zx - 3x^2) dx \\ &= \frac{z^3}{8} + \frac{3}{2} z - 1 - \frac{3}{8} z^3 + \frac{z^3}{8} = \frac{3}{2} z - 1 - \frac{1}{8} z^3 \end{aligned}$$

当 $z \geq 2$ 时

$$F_z(z) = 1$$

综合得:

$$F_Z(z) = \begin{cases} 0 & (z < 0) \\ \frac{3}{8} z^3 & (0 \leq z < 1) \\ \frac{3}{2} z - 1 - \frac{1}{8} z^3 & (1 \leq z < 2) \\ 1 & (z \geq 2) \end{cases}$$

所以密度函数为:

$$f_z(z) = F'_z(z) = \begin{cases} \frac{9}{8} z^2 & (0 \leq z < 1) \\ \frac{3}{2} - \frac{3}{8} z^2 & (1 \leq z < 2) \\ 0 & \text{其他} \end{cases}$$

5. 设 (X_1, Y_1) (X_2, Y_2) 的密度函数 $f_1(x, y), f_2(x, y)$ 分别为

$$f_1(x, y) = \begin{cases} k_1 e^{-3x-4y}, & x > 0, y > 0 \\ 0, & \text{其他} \end{cases}$$

$$f_2(x, y) = \begin{cases} k_2 e^{-3x-4y}, & x > y > 0 \\ 0, & \text{其他} \end{cases}$$

求:

- (a) 常数 k_1, k_2 ;
 (b) 边缘密度;
 (c) 条件密度函数;

解:

(a)

$$\begin{aligned}\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_1(x, y) dx dy &= \int_0^{\infty} \int_0^{+\infty} k_1 e^{-3x-4y} dx dy \\ &= k_1 \int_0^{+\infty} e^{-3x} dx \int_0^{+\infty} e^{-4y} dy = \frac{k_1}{12} = 1\end{aligned}$$

得 $k_1 = 12$

$$\begin{aligned}\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_2(x, y) dx dy &= \iint_b k_2 e^{-3x-4y} dx dy \\ k_2 \int_0^{+\infty} e^{-3x} dx \int_0^x e^{-4y} dy &= k_2 \int_0^{+\infty} e^{-3x} \cdot \frac{1}{-4} [e^{-4x} - 1] dx \\ &= -\frac{k_2}{4} \left[\int_0^{+\infty} e^{-7x} dx - \int_0^{+\infty} e^{-3x} dx \right] \\ &= \frac{k_2}{21} = 1\end{aligned}$$

得 $k_2 = 21$

$$\begin{aligned}f_1(x, y) &= \begin{cases} 12e^{-3x-4y}, & x > 0, y > 0 \\ 0, & \text{其他} \end{cases} \\ f_2(x, y) &= \begin{cases} 21e^{-3x-4y}, & x > y > 0 \\ 0, & \text{其他} \end{cases}\end{aligned}$$

(b)

$$f_{x_1}(x) = \int_{-\infty}^{\infty} f_1(x, y) dy$$

当 $x < 0$ 时, $f_1(x, y) = 0$ 故 $f_{x_1}(x) = 0$

当 $x \geq 0$ 时

$$\begin{aligned}f_1(x, y) &= \begin{cases} 12e^{-3x-4y}, & y > 0 \\ 0, & \text{其他} \end{cases} \\ f_{x_1}(x) &= \int_0^{\infty} 12e^{-3x-4y} dy = 3e^{-3x} \\ f_{x_1}(x) &= \begin{cases} 3e^{-3x}, & x \geq 0 \\ 0, & x < 0 \end{cases}\end{aligned}$$

同理

$$f_{Y_1}(y) = \begin{cases} 4e^{-4y}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

当 $x < 0$ 时, $f_2(x, y) = 0$ 故 $f_{x_2}(x) = 0$

当 $x \geq 0$ 时

$$f_2(x, y) = \begin{cases} 21e^{-3x-4y}, & 0 \leq y \leq x \\ 0, & \text{其他} \end{cases}$$

故:

$$f_{x_2}(x) = \int_0^x 21e^{-3x-4y} dy = \frac{21}{4} [e^{-3x} - e^{-7x}]$$

$$f_{x_2}(x) = \begin{cases} \frac{21}{4} [e^{-3x} - e^{-7x}], & x \geq 0 \\ 0, & \text{其他} \end{cases}$$

当 $y < 0$ 时, $f_2(x, y) = 0$ 故 $f_{Y_2}(y) = 0$

当 $y \geq 0$ 时

$$f_2(x, y) = \begin{cases} 21e^{-3x-4y}, & y \leq x \\ 0, & \text{其他} \end{cases}$$

$$f_{Y_2}(y) = \int_y^{+\infty} 21e^{-3x-4y} dx = 7e^{-7y}$$

$$f_{Y_2}(y) = \begin{cases} 7e^{-7y}, & y \geq 0 \\ 0, & \text{其他} \end{cases}$$

(c) 当 $y \geq 0$ 时

$$f_{x_1|Y_1}(x|y) = \frac{f_1(x, y)}{f_{Y_1}(y)} = \begin{cases} 3e^{-3x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

当 $y < 0$ 时

$$f_{x_1|Y_1}(x|y) = 0, \quad x \in (-\infty, +\infty)$$

当 $x \geq 0$ 时

$$f_{Y_1|X_1}(y|x) = \frac{f_1(x, y)}{f_{X_1}(x)} = \begin{cases} 4e^{-4y}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

当 $x < 0$ 时

$$f_{Y_1|X_1}(y|x) = 0, y \in (-\infty, +\infty)$$

当 $y \geq 0$ 时

$$f_{x_2/Y_2}(x|y) = \frac{f_2(x, y)}{f_{Y_2}(y)} = \begin{cases} 3e^{-3x+3y}, & x \geq y \\ 0, & x < y \end{cases}$$

当 $y < 0$ 时

$$f_{x_2/Y_2}(x, y) = 0, \quad x \in (-\infty, +\infty)$$

当 $x \geq 0$ 时

$$f_{Y_2|X_2}(y|x) = \frac{f_2(x, y)}{f_{X_2}(x)} = \begin{cases} \frac{4e^{-4y}}{1-e^{-4x}}, & 0 < y \leq x \\ 0, & \text{其他} \end{cases}$$

当 $x < 0$ 时

$$f_{x_2|X_2}(y|x) = 0, \quad y \in (-\infty, +\infty)$$

6. 一仪器由两个部件组成, 分别以 X, Y 表示这两个部件的寿命 (小时)。已知 (X, Y) 的分布函数

$$F(x, y) = \begin{cases} 1 - e^{-0.01x} - e^{-0.01y} + e^{-0.01(x+y)}, & x > 0, y > 0 \\ 0, & \text{其他} \end{cases}$$

试求:

(a) 边缘分布函数 $F_X(x), F_Y(y)$;

(b) $P\{1 < X \leq 2, 1 < Y \leq 2\}$;

(c) 密度函数 $f(x, y)$

解:

(a) 由边缘分布函数的定义知:

$$F_X(x) = F(x, +\infty) = \begin{cases} 1 - e^{-0.01x}, & x > 0 \\ 0, & \text{其他} \end{cases}$$

$$F_Y(y) = F(+\infty, y) = \begin{cases} 1 - e^{-0.01y}, & y > 0 \\ 0, & \text{其他} \end{cases}$$

(b) 由联合分布函数的定义知

$$P\{1 < X \leq 2, 1 < Y \leq 2\} = F(2, 2) - F(2, 1) - F(1, 2) + F(1, 1) = e^{-0.02} - 2e^{-0.03} + e^{-0.04}$$

(c)

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y} = \begin{cases} (0.01)^2 e^{-0.01(x+y)}, & x > 0, y > 0 \\ 0, & \text{其他} \end{cases}$$

7. 设二维随机变量 (X, Y) 服从矩形区域 $D = \{(x, y) | 0 \leq x \leq 2, 0 \leq y \leq 1\}$ 的均匀分布, 且

$$U = \begin{cases} 0, & X \leq Y \\ 1, & X > Y \end{cases}, V = \begin{cases} 0, & X \leq 2Y \\ 1, & X > 2Y \end{cases}$$

求 U 与 V 的联合概率分布

解: 由题意知, U, V 只能取 0 和 1, 故 (U, V) 的概率分布为

$$P = \{U = 0, V = 0\} = P\{X \leq Y, X \leq 2Y\} = P\{X \leq Y\} = \int_0^1 dx \int_x^1 \frac{1}{2} dy = \frac{1}{4}$$

$$P = \{U = 0, V = 1\} = P\{X \leq Y, X > 2Y\} = P\{X \leq Y\} = 0$$

$$P = \{U = 1, V = 0\} = P\{X > Y, X \leq 2Y\} = P\{X \leq Y\} = \int_0^1 dy \int_y^{2y} \frac{1}{2} dx = \frac{1}{4}$$

$$P = \{U = 1, V = 1\} = 1 - P\{X \leq Y, X \leq 2Y\} - P\{X \leq Y, X > 2Y\} - P\{X > Y, X \leq 2Y\} = \frac{1}{2}$$

8. 设二维随机变量 (X, Y) 在区域 $D = \{(x, y) | 0 < x < 1, x^2 < y < \sqrt{x}\}$ 上服从均匀分布, 令

$$U = \begin{cases} 1, & X \leq Y \\ 0, & X > Y \end{cases}$$

- (a) 写出 (X, Y) 的概率密度;
 (b) 问 U 与 X 是否相互独立? 并说明理由;
 (c) 求 $Z = U + X$ 的分布函数

解:

- (a) 根据题意, 区域 D 的面积为 $S_D = \int_0^1 (\int_{x^2}^{\sqrt{x}} dy) dx = \int_0^1 (\sqrt{x} - x^2) dx = \frac{1}{3}$, 则 (X, Y) 的概率密度为

$$f(x, y) = \begin{cases} 3, & (x, y) \in D \\ 0, & \text{其他} \end{cases}$$

- (b) 对于 $0 < t < 1$, 有

$$P(U \leq 0, X \leq t) = P(X > Y, X \leq t) = \int_0^t dx \int_{x^2}^x 3 dy = \frac{3}{2} t^2 - t^3$$

$$P(U \leq 0) = P(X > Y) = \frac{1}{2}$$

$$P(X \leq t) = \int_0^t dx \int_{x^2}^{\sqrt{x}} 3 dy = 2t^{\frac{3}{2}} - t^3$$

由于 $P(U \leq 0, X \leq t) \neq P(U \leq 0)P(X \leq t)$, 所以 U 与 X 不相互独立

- (c) 当 $z < 0$ 时, $F(z) = 0$ 当 $0 \leq z < 1$ 时, 有

$$F(z) = P(U + X \leq z) = P(U = 0, X \leq z) = P(X > Y, X \leq z) = \frac{3}{2} z^2 - z^3$$

当 $1 \leq z < 2$ 时, 有

$$F(z) = P(U + X \leq z) = P(U = 0, X \leq z) + P(U = 1, X \leq z - 1) = \frac{1}{2} + 2(z - 1)^{\frac{3}{2}} - \frac{3}{2}(z - 1)^2$$

当 $z \geq 2$ 时, 有

$$F(z) = P(U + X \leq z) = 1$$

所以 $Z = U + X$ 的分布函数为

$$F(z) = \begin{cases} 0, & z < 0 \\ \frac{3}{2} z^2 - z^3, & 0 \leq z < 1 \\ \frac{1}{2} + 2(z - 1)^{\frac{3}{2}} - \frac{3}{2}(z - 1)^2, & 1 \leq z < 2 \\ 1, & z \geq 2 \end{cases}$$